Benefits of CubeSats for scientific investigations of the upper atmosphere

Ennio Sanchez, Michael Nicolls, Russell Cosgrove, and Hasan Bahcivan

Center for Geospace Studies, SRI International, Menlo Park, CA†

email: ennio.sanchez@sri.com

Abstract

Because of their low price, CubeSats are ideal for distributed measurements, low-altitude (short-lived) missions, highly specialized missions, and novel speculative missions, typically unfeasible for larger satellites costing orders of magnitude more. There is the potential for missions utilizing essentially standard instrumentation, which would nevertheless make novel scientific contributions: for example, an array of GPS occultation receivers could be used for the purpose of achieving along-track resolved occultation imaging; or a vector accelerometer could be sent on a short-lived mission (lasting several months), to detect waves, tides, and turbulence in the lower thermosphere. Other scientific goals may require innovative miniaturization efforts, but with high scientific payoff: for example, dense clusters of electric field probes could be used to investigate ionospheric phenomena, such as westward traveling surges, the development of storm-enhanced density plumes, and the initiation of equatorial spread F, that exhibit a range of temporal and spatial scales difficult to probe with single-satellite or ground-based measurements. Here, we use these examples to make the point that **CubeSats can enable a new kind of satellite-based research**, which stresses global (including low-altitude) coverage of space, and involvement of a broader community of scientists and educators pursuing miniaturization and novel mission formulation.

1 In-Situ Probing of the Lower Thermosphere (LT)

Neutral Dynamics Science

- Propagating and breaking waves are thought to be critical components of the lower thermosphere (80-300 km), producing shears, instabilities, secondary waves, and turbulence (Figure 1).
- There is very little direct observation. Our understanding is mostly derived from theoretical investigations, or by inference from observations at much lower altitudes.

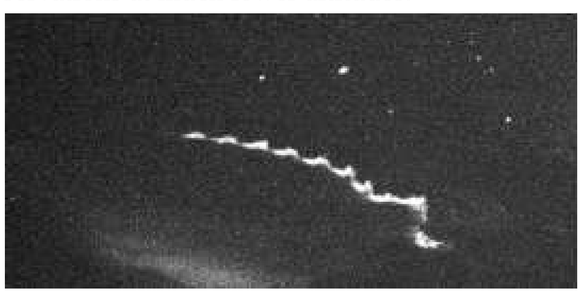

Figure 1: Atmospheric turbulence at ∼110 km, observed by a rocket chemical release. Reproduced from *Larsen et al.* [6].

Spaceborne Observations

- Region difficult to study directly with ground-based instruments (e.g., lidars generally reach up to ∼100 km).
- Probing generally done with rockets (one-shot deals) and radars (involving unknown electrodynamics).

CubeSat Experiment

- In absence of frictional forces, satellite orbits are completely free of acceleration (equivalence principle); accelerometers provide direct measure of frictional drag in the LT.
- Orbital lifetime in LT too short for dedicated large sat missions.
- CubeSats could make *in-situ* measurements in LT for ∼months.
- Cost-to-benefit tradeoff makes sense for small sats.
- CHAMP and GRACE use of accelerometers to infer neutral density at ∼400-500 km (e.g., Fig. 2) provides proof of concept.
- In LT, neutral density variations readily detectable using small accelerometers capable of being packaged in a CubeSat.
- Attitude determination required only for winds, and could use the accelerometer.

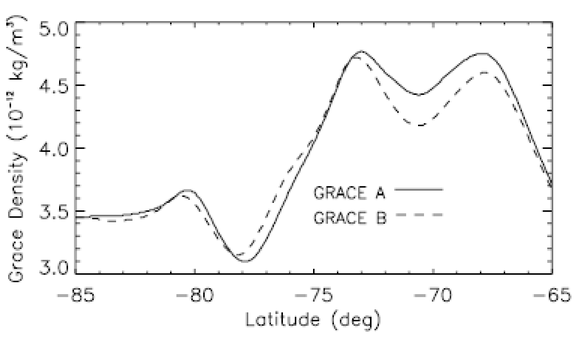

Figure 2: Large-scale wave structure observed by the GRACE accelerometer at ∼500 km. From *Bruinsma et al.* [3]

2 Occultation Imaging of the E-Region

E-Region Science

- Off-equatorial E-region conductivity (with/without layers) essential for PRE, strongly linked to ESF occurrence [5].
- Narrow, transient ion layers (Fig. 3, left panel) exhibit a broad range of spatial and temporal scales - not very well understood.
- Layers possibly important in generation of equatorial spread F (ESF) through: modulation of off-equatorial conductivity, latitudinal gradient in Hall conductivity, large polarization electric fields, affecting F-region velocity shear magnitudes, etc.

Spaceborne Observations

- Most observations made from ground with ISRs or ionosondes; cannot provide necessary spatial information.
- Low densities (10^3-10^4 cm^{-3}) of nighttime E-region can only be reliably measured by ISRs (Arecibo, Jicamarca).
- Satellites (COSMIC, CHAMP) use GPS occultation to identify E-region structures [8], some have tried to resolve in altitude [1].

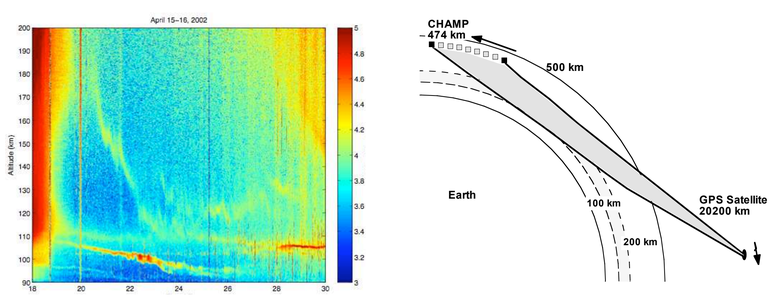

Figure 3: (Left) Log10 density profile measured at Arecibo on April 15-16, 2002. (Right) Vertical spatial probing of the ionosphere by the CHAMP GPS occultation experiment. Reproduced from *Bishop et al.* [1]

CubeSat Experiment

- Typical occultation geometry in Fig. 3 (right). Satellite measures TEC vs. time (altitude), horizontal distances 10s-100s km.
- Can isolate structures but difficult to reconstruct layers in both vertical and along-track dimensions - typically assume spatial homogeneity, spherical symmetry [4].
- Difficult to quantify accuracy of resulting profiles [1].
- String of CubeSats could improve along-track resolution - a limiting factor in the inversions.
- Cluster of CubeSats with spatially varying ionospheric pierce points allow imaging of layers, true measurement of dimensions.
- If able to extract 2-3D densities, compute conductivities & gradients - explore E-region contribution to ESF seeding/variability.
- Current generation occultation receivers are CubeSat-sized (IGOR receiver on COSMIC - 20x24x15 cm). Next generation will be smaller. Some attitude information may be required.

3 The Aurora: A Multi-Scale Problem

Auroral Science

- Discrete auroral structures observed at meter to hundreds of km spatial scales (Fig. 4), sub-minute to hour temporal scales.
- Auroral intensity, current, and convection self-similar from seconds to hours, meters to 1000s of km.
- Breakdown in fractality at through substorms - global energy release events with characteristic scales 0.5-1 hr, 100s km.
- To understand how electromagnetic M-I coupling evolves between equilibrium states, need to measure fields and particles in auroral structures - may exist on small/short scales (e.g., Fig. 5).

Spaceborne Observations

- Large amplitude (∼1 V/m), small scale (∼10-100 m) E-fields are seen by satellites in the topside ionosphere (Fig. 4).
- Such structures not seen by radars before because they are integrated out in time and/or space.
- Single spacecraft have seen snapshots of these features, but:
  (a)Do fields map to lower ionosphere or caused locally?
  (b)What is arc width distribution? How do field & particle populations change with time?

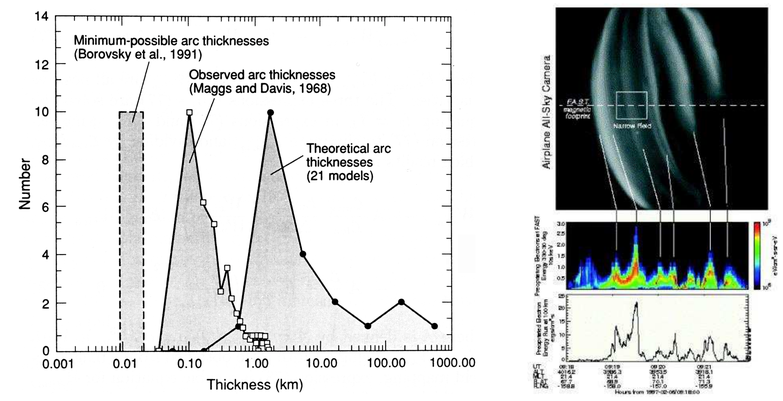

Figure 4: (Left) Distribution of observed and predicted arc thicknesses (from *Borovsky* [2]). (Right) FAST pass through multiple arc system viewed from an all-sky imager. Center panel is electron energy-time spectrogram. Bottom panel shows precipitating energy flux on a linear scale (Courtesy FAST Science Team).

CubeSat Experiment

- String of closely spaced CubeSats in high-inclination orbits, initial altitude ∼300 km, will provide necessary time resolution for field & particle measurements across small-scale arcs.
- Inter-spacecraft separation of ∼10 meters would allow sampling of same arc at 1-2 ms cadence.
- As atmospheric drag reduces orbits altitude, spacecrafts sampling region will include E-region.
- Another string of CubeSats deployed at ∼800 km altitude can stay in orbit for much longer period and sample topside ionosphere at near-conjugate points to low-altitude string - allows for measurement of E-field mapping efficiency.
- Current boom technology not applicable to CubeSat mass and volume constraints. Alternative approach required.

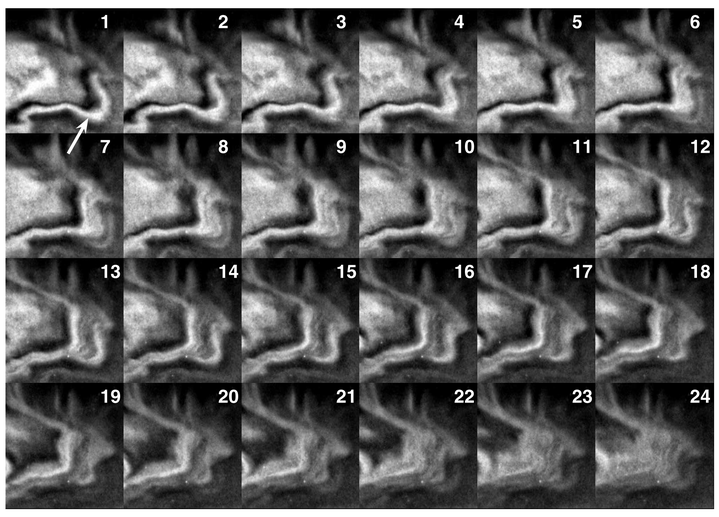

Figure 5: Image sequence showing cascading into multiple arc system. Sampling period is 20 ms, total time is 0.5 s (from *Semeter and Blixt* [7]).

4 Conclusions

We have outlined three mission concepts well suited to CubeSats:

- In-situ observations of lower thermospheric waves. Cost-benefit tradeoff untenable for larger spacecraft, and technology is readily available to be applied to CubeSats.
- Improved resolution in reconstruction of E-region ion layers and densities. Required spatial coverage can only be provided by large number of CubeSats. Receiver and attitude determination technology can be quickly adapted to CubeSats.
- CubeSat constellations will resolve electric fields of multi-scale phenomena. New measurement methods required. Opportunity for development of next generation of E-field probes.

The CubeSat paradigm allows satellite research involving a broad community of scientists and educators, pursuing miniaturization and novel mission formulation.

References

[1] R. L. Bishop, V. Wong, M. J. Nicolls, P. Strauss, M. C. Kelley, and N. Aponte. Comparison of nighttime E-region density profiles obtained from the Arecibo Observatory and GPS occultation measurements. *J. Geophys. Res.*, 2007.

[2] J. Borovsky. Auroral arc thickness as predicted by various theories. *J. Geophys. Res.*, 98:6101, 1993.

[3] S. Bruinsma, J. M. Forbes, R.S. Nerem, and X. Zhang. Thermospheric density response to the 20-21 November 2003 solar and geomagnetic storm from Champ and Grace accelerometer data. *J. Geophys. Res.*, 111:A06303, doi:10.1029/2005JA011284, 2006.

[4] G. A. Hajj, E. R. Kursinsky, L. J. Romans, W. I. Bertiger, and S. S. Leroy. A technical description of atmospheric sounding by GPS occultation. *J. Atmos. Solar-Terr. Phys*, 64:451–469, 2002.

[5] M. C. Kelley, V. K. Wong, G. A. Hajj, and A. J. Mannucci. On measuring the off-equatorial conductivity before and during convective ionospheric storms. *Geophys. Res. Lett.*, 31:L17805, doi:10.1029/2004GL020423, 2004.

[6] M. F. Larsen, M. Yamamoto, S. Fukao, R. T. Tsunoda, and A. Saito. Observations of neutral winds, wind shears, and wave structure during a sporadic-E/QP event. *Ann. Geophys.*, 23:2369–2375, 2005.

[7] J. Semeter and E. M. Blixt. Evidence for Alfvén wave dispersion identified in high-resolution auroral imagery. *Geophys. Res. Lett.*, 33:L13106, doi:10.1029/2006GL026274, 2006.

[8] D. L. Wu, C. O. Ao, G. A. Hajj, M. Juarez, and A. J. Mannucci. Sporadic E morphology from GPS-CHAMP radio occultation. *J. Geophys. Res.*, 110:A01306, doi:10.1029/2004JA010701, 2005.

†Address for correspondence: Center for Geospace Studies, SRI International, 333 Ravenswood Ave., Menlo Park, CA 94025